FlexTome: a 3D-printed, flexure-pivot rocking microtome with a differential-screw feed for 5 µm paraffin histology at a US$61.97 bill of materials

Design report: Engineering Mechanics / Biomedical Engineering

Abstract

Histopathology is the reference standard for cancer diagnosis, but much of sub-Saharan Africa has roughly one pathologist per million people and district hospitals lack the US$150 000–200 000 of equipment an essential pathology laboratory requires. The microtome that cuts 3–5 µm paraffin sections is one of the few items in that chain with no low-cost substitute: new manual rotary microtomes list at US$9 000–22 700, and the thinnest published open-source sectioning device cuts at ≥50 µm, an order of magnitude too thick for diagnostic H&E. We present FlexTome, a rocking-arm microtome designed to be printed on a 235 mm desktop FDM printer and completed with hardware-store fasteners. Three design decisions make 5 µm sectioning plausible from printed plastic: (i) the arm pivot is a cross-strip flexure built from three 0.20 mm spring-steel feeler-gauge strips, so the metrology loop contains no sliding or rolling joint and therefore no backlash; (ii) the 5 µm feed is generated by a differential screw (M8×1.25 against M8×1.0, 0.25 mm per revolution) indexed by a 50-tooth printed ratchet that the arm itself advances at the top of each stroke, so the micrometre-scale motion is carried entirely by steel threads while the printed parts need only millimetre tolerances; and (iii) the disposable blade rides on a monolithic printed parallelogram flexure stage preloaded against the screw. A crank-rocker drive gives a 54 mm arc stroke at ±12.5°. An analytic structural-loop model, calibrated by plane-stress finite-element models of the flexure stage, of the frame-nut upstand and of the frame, predicts a feed-axis loop stiffness of 0.83 N/µm (1.20 µm/N). Two modelling lessons shaped the design: the frame model caught a 0.83 µm/N tower compliance that a beam model had missed and drove a gusset revision that cut it to 0.21 µm/N; and an external review of the first release showed that its feed-screw upstand, blade clamp and coupler did not form the load path the budget assumed (the upstand alone was 7.3 µm/N), which led to the connected, gusseted revision analysed here. A Monte-Carlo budget that treats every error as a stroke-to-stroke difference, with correlated thread-lead error, cutting-force variation, printed-ratchet error, pawl seating and thermal drift, predicts a section thickness of 5.0 ± 0.7 µm (1 s.d.; 90 % interval 3.8–6.2 µm) at the 5 µm setting with a thread-lead bias within ±0.7 %, comparable to the > ±10 % deviation measured on commercial rotary microtomes in a 154-site survey. All structural resonances (≥357 Hz) lie far above the <9 Hz cutting excitation, so chatter is not predicted. The bill of materials is US$61.97, a > 99 % reduction relative to new list price. These are model predictions: the machine has not yet been built or measured. The design, analysis code and a validation protocol are released so that the predictions can be tested on a physical build.

1 Introduction

1.1 The problem

The cancer-diagnosis chain in low- and middle-income countries (LMICs) fails at the laboratory before it fails at the clinic. A 2016 survey of African pathology resources found population-per-pathologist ratios from 84 000 (Mauritius) to 9.3 million (Niger), against 15 000–20 000 in the UK and US [1]; the Lancet Commission on diagnostics estimates that 47 % of the world's population has little or no access to diagnostics [2]; and an essential district-hospital pathology package costs US$150 000–200 000 in equipment [3]. Within that package, the microtome is the only instrument that converts a fixed, paraffin-embedded biopsy into the 3–5 µm section a pathologist can read.

New manual rotary microtomes list at US$22 690 (Leica RM2125 RTS), US$21 548 (Sakura Accu-Cut SRM 200) and US$8 995 (Thermo HM 325); used units sell for US$2 100–7 950 (vendor prices, October 2026).

1.2 Why no low-cost microtome exists

Open-source hardware has reached the microscope: the OpenFlexure microscope is printed for US$20–200 and has been manufactured locally in Tanzania and Kenya [4]. It has not reached the microtome. The published low-cost sectioning devices cut at ≥50 µm (a hair-clipper vibrating razor [5]), <100 µm for hand-fed plant tissue (Rapid-Tome [6]) and 250 µm–1 mm for live organoids (a US$853 HardwareX slicer [7]). None embeds in paraffin, none uses a standard disposable microtome blade, and none has a micrometre-resolution feed. The barrier is mechanical: a 5 µm section demands that the knife-to-specimen distance repeat to better than ∼1 µm stroke after stroke, while printed plastic parts have ±0.1–0.2 mm dimensional scatter [8], creep under sustained load [9], a Young's modulus 60 times lower than steel [10], and sliding joints printed in plastic have backlash of tens of micrometres.

1.3 The design thesis

Our thesis is that these limitations can be designed around rather than overcome, using three principles from precision-instrument design:

1. **No sliding joints in the metrology loop.** Every joint between the blade edge and the specimen face that must repeat to <1 µm is a flexure: the arm pivot is a cross-strip flexure, the blade stage is a parallelogram flexure, and the screw-to-stage interface is a preloaded nut. Flexures have zero backlash and no stick-slip [11, 12].
2. **Keep the micrometre chain in steel.** The feed increment (5 µm) is set by the pitch difference of two standard steel threads and the tooth count of a ratchet; the printed parts only hold those steel parts in roughly the right place. The differential screw turns a 0.25 mm pitch difference into fine motion without a precision thread [13].
3. **Make the structural loop short and deep.** Section-to-section thickness variation equals loop compliance × cutting-force variation; chatter is a stiffness phenomenon [14, 15]. The loop is a one-piece ribbed base and a gusseted tower whose root depth (60 mm) was set by finite-element analysis.

The rocking-arm kinematics themselves are not new: Horace Darwin's 1885 Cambridge rocker used a pivoted arm, screw feed and ratchet, and a modified rocker later cut sections for electron microscopy [16]. What is new is the realisation of that kinematic in printed plastic and commodity steel at a US$61.97 bill of materials, with an explicit, FE-checked stiffness and error budget.

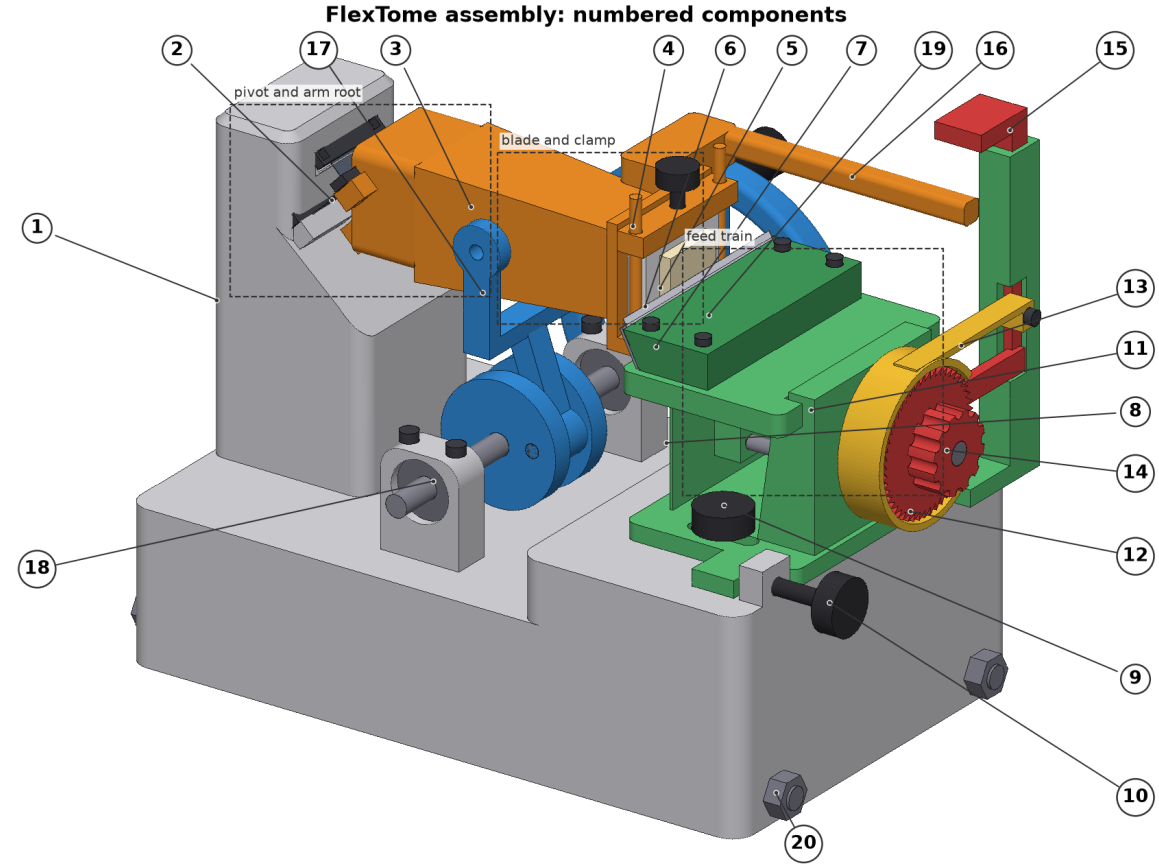
2 Design

2.1 Architecture

Figure 1 shows the assembly. The coordinate frame has x along the feed (specimen → blade), y along the blade edge and z vertical. Table 1 summarises the subsystems.

2.2 Kinematics

The crank-rocker (Fig. 2a,b; Sec. 3) gives $\theta = \pm 12.5°$, a block-face stroke of 54 mm and a peak cutting speed of 174 mm/s at 60 rpm, one section per handwheel turn. Because the specimen travels on an arc, the face retracts 3.0 mm from the blade at the stroke ends and the cut face is a cylinder of radius 125 mm (sagitta 0.40 mm over a 20 mm block). The latter is the rocker's classical "curved section" property; because every stroke follows the identical arc, the curvature is cut into the block face once and the next section (the difference between two successive arcs) has uniform thickness. The effective clearance angle varies by ±4.6° across a 20 mm face; a 27° blade lean



1	Frame: ribbed base 235 × 150 × 55 mm, gusseted tower (one print)	**11**	Differential feed screw: M8×1.0 bolt, coupler, M8×1.25 rod
2	Cross-strip flexure pivot: 3 × 0.20 mm steel strips in clamp pads	**12**	Ratchet drum 50 T × 14 mm (5 µm per tooth) with jam nuts
3	Rocking arm: tapered hollow box beam, 125 mm pivot to face	**13**	Thickness-selector shield (1 to 6 teeth = 5 to 30 µm)
4	Cassette clamp: lower lip, sliding jaw, M5 thumbscrew	**14**	Feed knob
5	Histology cassette and paraffin block (consumable)	**15**	Pawl rod with finger, lifted by the striker bar at top of stroke
6	Disposable low-profile blade 80 × 8 × 0.25 mm, 27° lean	**16**	Striker bar on arm (feed actuation)
7	Blade holder (full-width back wall), 1.0 mm clamp strip, 2 × M4	**17**	Fork connecting rod 71.2 mm and crank webs, r = 13 mm
8	Parallelogram flexure stage (one print), plates 1.0 × 44 mm	**18**	Built-up crankshaft, 8 mm stubs in 2 × 608 bearings
9	Stage clamp knobs M6 (coarse-approach slots ±10 mm)	**19**	Handwheel 110 mm with handle (one turn = one section)
10	Coarse-approach screw M6 in side boss	**20**	M8 tie rods (base pre-compression, split-base option)

Figure 1: FlexTome assembly with numbered components (frame footprint 235 × 150 mm; overall envelope 272 × 210 × 202 mm including handwheel, feed knob and pawl rod). The specimen in a standard histology cassette (5) is carried on a 125 mm printed arm (3) that rocks ±12.5° on a cross-strip steel flexure pivot (2). The blade (6) sits on a parallelogram flexure stage (8) driven along x by a differential screw (11); a striker bar on the arm (16) lifts a pawl rod (15) at the top of each stroke, advancing the 50-tooth ratchet drum (12) by 1–6 teeth (5–30 µm) as set by the selector shield (13). A crank-rocker (17, 18) with 608 bearings and a 110 mm handwheel (19) drives the arm. Rendered from the parametric CAD model; Sec. 3 walks through each subsystem.

Table 1: Subsystems and their role in the error budget.

Subsystem	Implementation	Role in the error budget
Frame	One printed part: 235 × 150 × 55 mm ribbed base, 118 mm tower with a forward gusset (60 mm root depth tapering to 32 mm at the pivot head), raised front deck; two M8 steel tie rods through the base	Closes the structural loop (0.21 µm/N pivot-to-deck by FE with the base in full contact, 0.35 µm/N on two end pads; the latter is used); tie rods pre-compress the print against creep/warp and allow a split base for small printers
Arm pivot	Cross-strip flexure: three strips (10/20/10 mm wide) of 0.20 mm feeler-gauge steel, 20 mm free length, crossing at mid-length, clamped in printed pads with M3 screws	Zero backlash; axially stiff (40 N/µm); rotational stiffness 0.27 N m/rad
Arm	Tapered hollow box beam 40 → 30 mm, 125 mm pivot-to-face; integral cassette clamp (fixed lower lip, sliding jaw, M5 thumbscrew)	Loaded axially during the cut; 0.08 µm/N
Drive	Crank radius 13 mm at 60 mm from pivot, crank centre 71.2 mm below the arm joint; 71.2 mm fork connecting rod running between two printed crank webs; built-up crankshaft of two 8 mm stubs in two 608 bearings; 110 mm spoked handwheel	Kinematics only; not in the metrology loop
Blade stage	Monolithic printed parallelogram: two 1.0 × 44 mm plates (60 mm and 2 × 20 mm wide), platform, hanging nut tab; bolted to deck through slots (±10 mm coarse approach)	Guides the blade along x with no sliding; pitch stiffness carries the cutting moment
Blade holder	Disposable low-profile blade (80×8×0.25 mm) leaning back 27° from vertical, its rear face bearing on a full-width printed back wall; 1.0 mm steel clamp strip in front, pulled onto a 0.20 mm blade recess by two M4 screws below the blade; 2.5 mm edge overhang	The cutting force pushes the blade onto the back wall (not onto screw tips); clearance 9.5° nominal; overhang compliance 0.25 µm/N
Fine feed	Differential screw: M8×1.0 bolt in the stage-tab nut; its hex head sits in a printed coupler whose captured nut carries the M8×1.25 rod, screwed in until the rod end bears on the bolt head (steel-on-steel axial path); the rod runs in the frame nut of a gusseted upstand → 0.25 mm/rev stage motion while the screw itself translates 1.25 mm/rev; both nuts sit on 16 mm steel washers; two extension springs preload the tab onto one thread flank over the whole travel; 14 mm long 50-tooth ratchet drum keeps the pawl engaged over the ±5 mm screw travel; sector shield on the column selects 1–6 teeth; stage travel ±1 mm (400 sections at 5 µm) per reset	Sets the increment; thread-lead scatter 0.18 µm (1 s.d.) and bias within ±0.7 %; upstand 0.19 µm/N (FE)
Feed actuation	Vertical pawl rod in a printed guide, lifted ≤22 mm by a striker bar fixed to the arm during the last ∼6° of the upstroke; gravity/spring return	Feed occurs early in the down-stroke, when the striker releases the rod at $\theta \approx$ ang6, before the face reaches the edge

(clearance 9.5° at mid-face, rake 45.5° with a 35° blade facet) keeps it within 4.9–14.1°, inside the 3–15° range practitioners recommend (Fig. 2c).

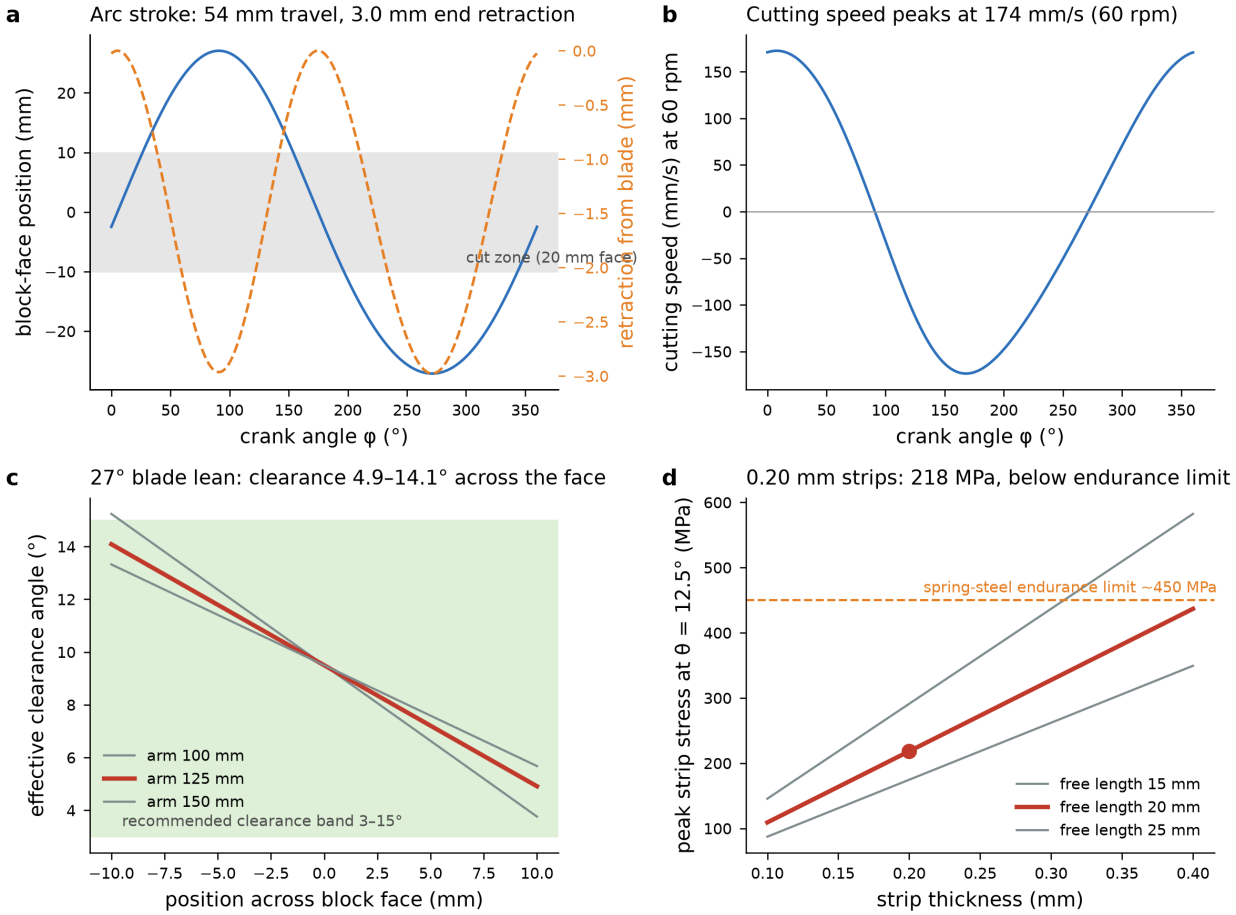


Figure 2: Kinematics and pivot. (a) Face position and retraction vs crank angle. (b) Cutting speed. (c) Effective clearance angle across the face for three arm lengths. (d) Peak pivot-strip bending stress vs strip thickness; the chosen 0.20×20 mm strips reach 218 MPa at full swing, below the ~450 MPa endurance limit of hardened spring steel (feeler-gauge stock; the exact limit must be taken from the stock datasheet). Rotational stiffness is 0.267 N m/rad (torque 0.058 N m at full swing); the second-order centre shift is $\approx 0.1\,L\theta^2$ ≈95 µm at ±12.5°, which is repeatable and therefore shapes the arc rather than the thickness [12, 17].

2.3 Feed resolution

With pitches $p_1 = 1.25$ mm and $p_2 = 1.00$ mm, one revolution advances the stage by $p_1 - p_2 = 0.25$ mm toward the specimen; one tooth of the 50-tooth ratchet is 5.0 µm. The selector shield exposes $k = 1 \ldots 6$ teeth to the pawl, giving 5, 10, 15, 20, 25 or 30 µm settings (trimming at 20–30 µm, sectioning at 5 µm). Only the *difference* of the two threads matters, so ordinary rolled-thread hardware is sufficient: the budget assumes 1 % independent local pitch error on each thread and 3 µm peak-to-peak periodic error per revolution.

2.4 Blade stage design

Figure 3a maps plate thickness and height against bending stress at the ±1 mm feed travel and against the x-stiffness that sets the preload. The chosen 1.0×44 mm plates give $k_x = 4.9$ N/mm (analytic) / 4.0 N/mm (FE), ≤5 N preload on the screw, 5.4 MPa peak plate stress and a creep safety factor of 2.8 against a 15 MPa sustained-stress limit for PLA [9]; PETG or ASA are drop-in alternatives. In the screw-held condition the stage is a short cantilever about the tab nut; with the gusseted tab of revision 2 the FE model (Fig. 4) gives 0.132 µm/N at the blade edge, and this

value is used in the budget. Because a symmetric parallelogram is neutral at mid-travel, its own restoring force would reverse direction across the ±1 mm range and let the nut change thread flank; two light extension springs (∼10 N total) between the tab and the upstand therefore keep the nut on the flank that the cutting force also loads.

A consequence of the differential principle that is easy to overlook: the screw body advances through the frame nut by the full 1.25 mm per revolution while the stage moves 0.25 mm, so over the ±1 mm stage range the screw, coupler, ratchet and knob translate ±5 mm. The ratchet is therefore a 14 mm drum rather than a wheel, the 13 mm coupler sits in a 24 mm gap between the stage tab and the upstand, and the thickness-selector shield is mounted on the stage body, not on the screw. A boolean interference audit of the assembly (Sec. 3.8) confirmed that no unintended contacts remain over the full stroke and screw travel.

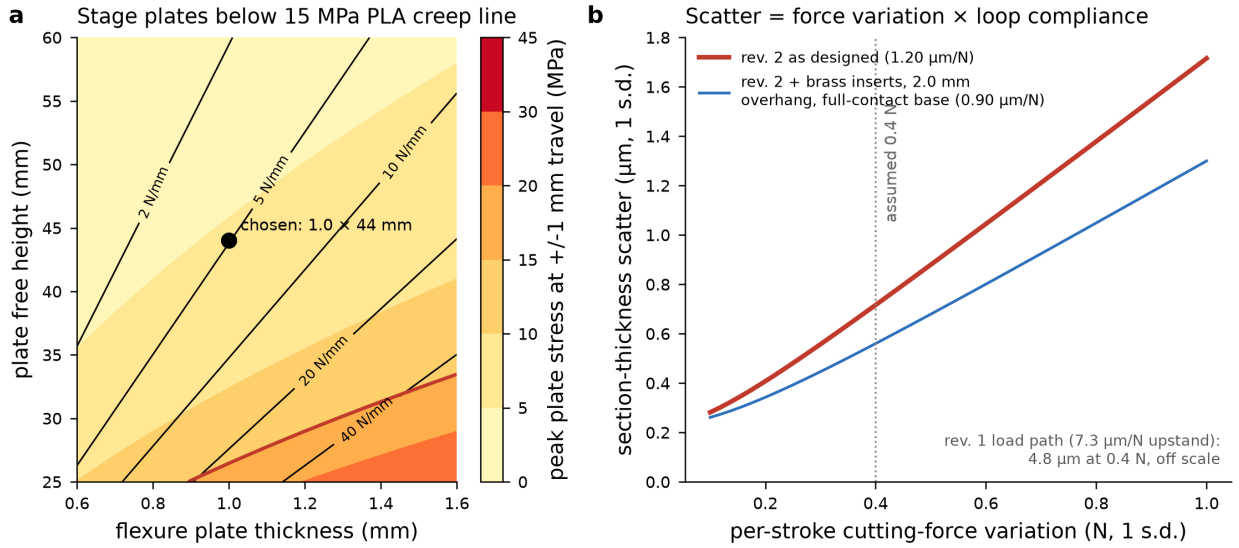


Figure 3: (a) Design map of the flexure stage; the red contour is the 15 MPa PLA creep line. (b) Predicted thickness scatter vs cutting-force variation for the original slender tower and the gusseted tower; the gusset reduces scatter by ∼25 % at the assumed 0.4 N force variation.

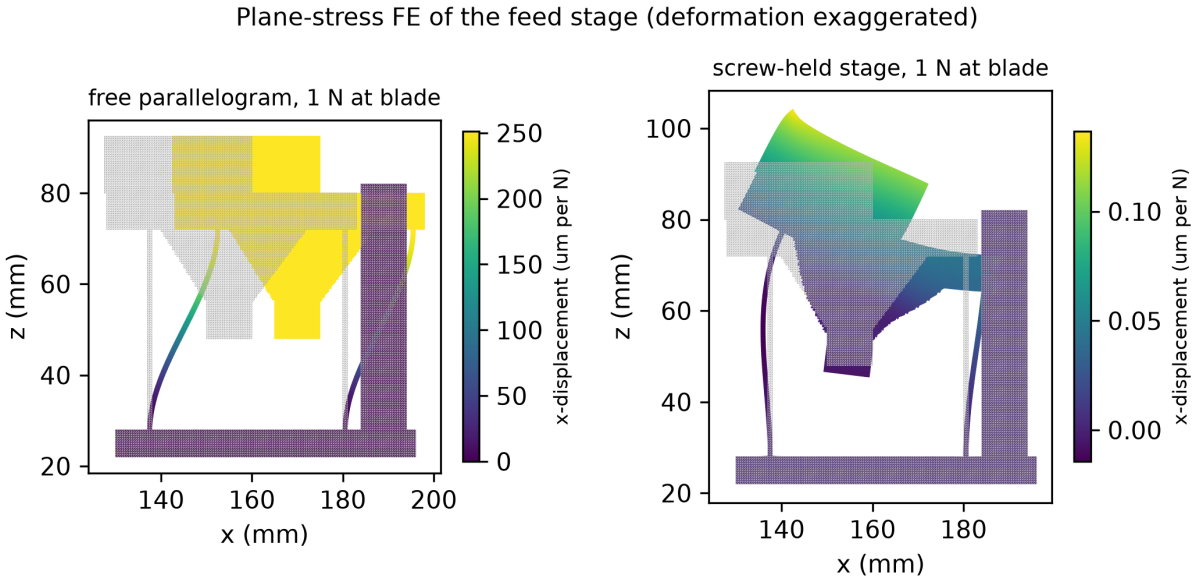


Figure 4: Plane-stress finite-element model of the revision-2 stage (scikit-fem, quadratic triangles, 16 882 elements; out-of-plane widths represented by scaling E per region). Left: free stage under 1 N at the blade, $k_x = 3.98$ N/mm. Right: stage held at the tab nut, compliance 0.132 µm/N at the blade.

2.5 Frame design: an FE-driven revision

The beam model treated the tower as a 70×70 mm box and the base as a plate rotating under the tower-root moment, and predicted that two M8 tie rods would triple base rigidity. A plane-stress FE model of the frame side-section (Fig. 5) showed otherwise: with the tower as actually drawn (32 mm deep), pivot-to-deck compliance was 0.83 µm/N, essentially all tower bending, and the tie rods changed it by only 2 % because the ribbed base was already stiff. Adding a forward gusset (the tower root widened to 60 mm over the first 55 mm of height, tapering to the 32 mm head, within the existing footprint and clear of the arm's swing) reduced the frame compliance to 0.21 µm/N with the whole underside bearing on the bench. Because a printed base may warp and bear only at its ends, the budget uses the two-end-pad support case, 0.35 µm/N (0.42 µm/N without the tie rods, which do matter in that case). The tie rods were retained for a different reason: they pre-compress the printed base against creep and warp and allow the base to be printed in two halves on small printers.

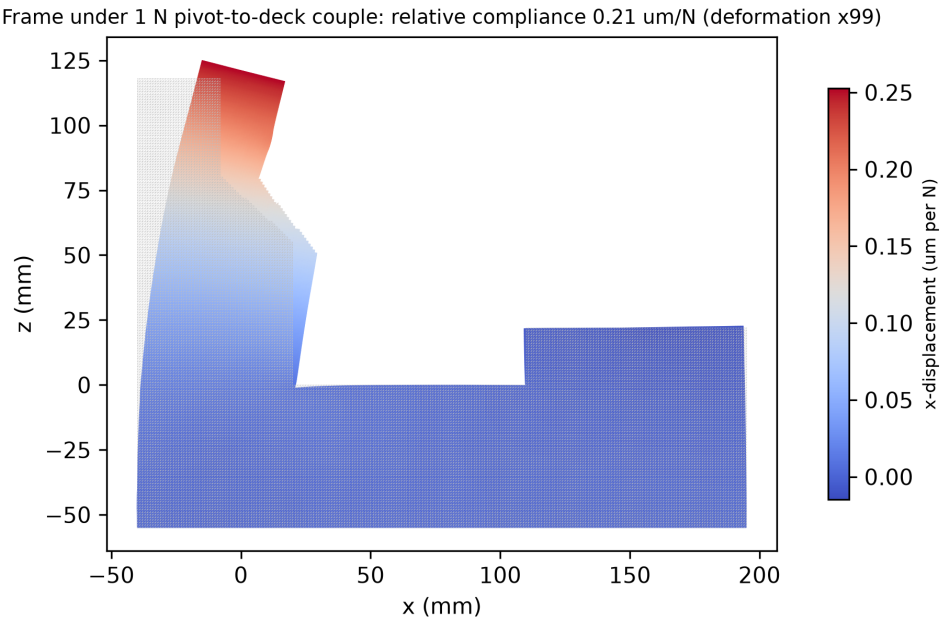
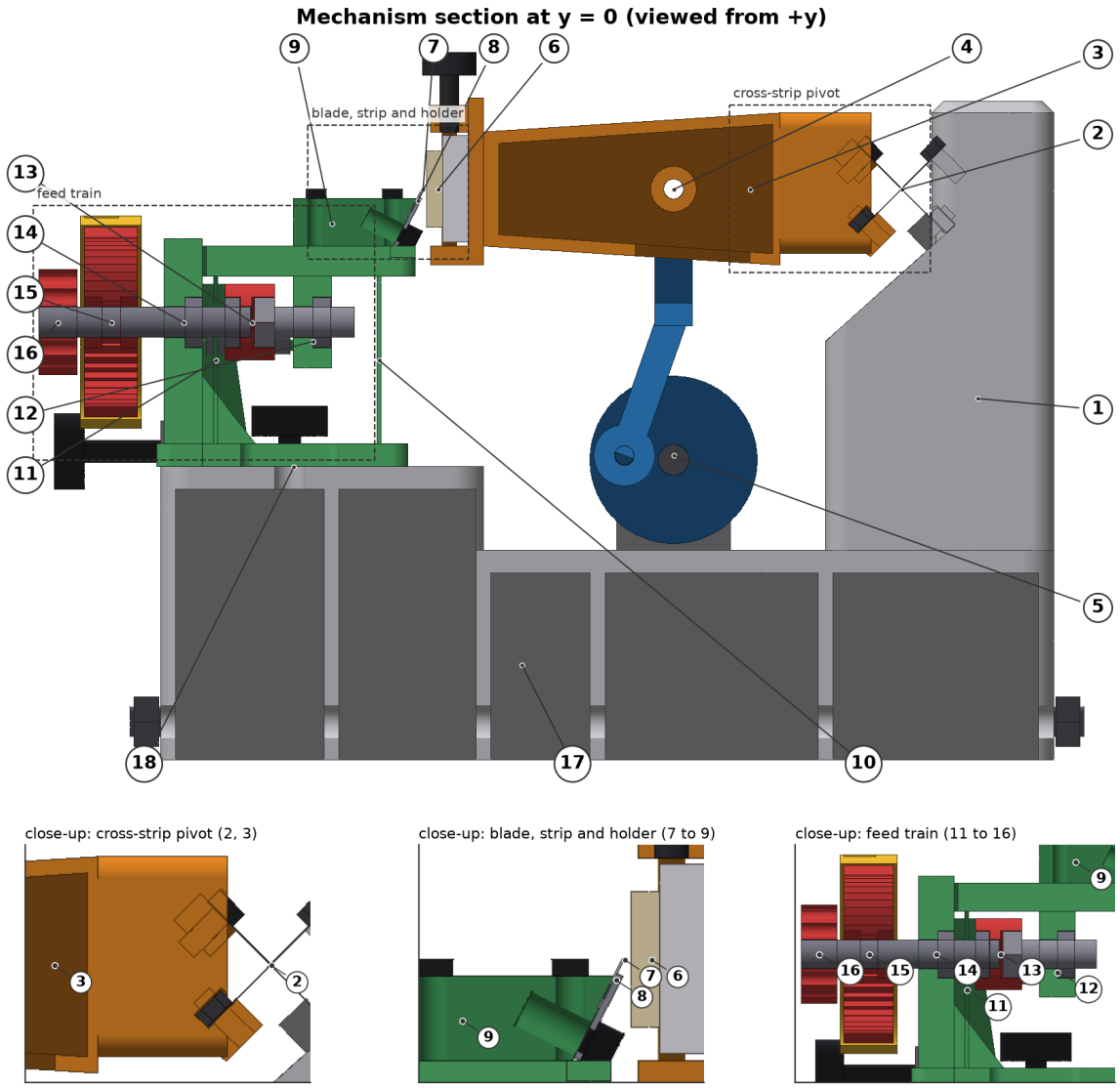


Figure 5: Plane-stress FE of the gusseted frame under a self-equilibrated 1 N couple between the pivot and the stage-foot clamp zone (quadratic triangles, 42 281 elements; ribbed base represented by a 40 % web fraction with solid 6 mm skins). Relative pivot-to-deck compliance 0.21 µm/N (full-contact base; 0.35 µm/N on two end pads), versus 0.83 µm/N for the original slender tower.

3 Mechanism walkthrough

This section follows the machine from the handwheel to the section ribbon, naming every part (Fig. 1, numbered 1–20; Fig. 6, cut at $y = 0$) and giving the governing relation for each stage with the numbers used in the design. Throughout, x is the feed axis (specimen → blade), y runs along the blade edge, z is up, and the pivot axis is the y-parallel line through $P = (0, 0, 95)$ mm.



1 Gusseted tower: 60 mm root depth, FE compliance 0.21 µm/N
2 Cross-strip pivot: rotation about the strip crossing, zero backlash
3 Arm root clamp pads (M3) and hollow box beam
4 Rod joint at r = 60 mm; crank pin r = 13 mm gives θ = ±12.5°
5 Crank webs (pin between them), shaft axis z = 23.8 mm
6 Cassette and block: face on 125 mm arc, 54 mm stroke
7 Blade edge at (126, 95): lean 27°, clearance 9.5°, rake 45.5°
8 Clamp strip top 2.5 mm below the edge, outside the swept circle
9 Blade holder: full-width back wall carries the cutting force
10 Front flexure plate 1.0 × 44 mm (60 mm wide)
11 Rear flexure plates 2 × 20 mm wide
12 Gusseted stage tab, M8×1.0 nut on washer (moves ±1 mm)
13 Coupler: bolt head in pocket, rod end bears on it; travels ±5 mm
14 Frame nut M8×1.25 on washer in gusseted upstand (FE 0.19 µm/N)
15 Ratchet drum 14 mm long: pawl stays engaged over screw travel
16 Feed knob
17 Rib pockets (open underside) and M8 tie rod at z = −45 mm
18 Raised deck (z = 22 mm): stage clamp slots and coarse approach

Figure 6: Section through the mid-plane $y = 0$, viewed from $+y$ (handwheel removed). The metrology loop runs from the blade edge (7) through the holder's back wall and platform (9), the flexure plates (10, 11), the gusseted stage tab and its nut (12), the fine bolt, coupler and rod (13), the frame nut in the gusseted upstand (14), the deck and base (17, 18), the tower (1), the cross-strip pivot (2) and the arm (3) to the block face (6). The drive (crank webs (5), connecting rod (4)) hangs below the arm and is outside the loop.

3.1 Drive train: handwheel to arm

The handwheel (19; 110 mm, five spokes, offset handle) turns a built-up crankshaft (18): two 8 mm steel shaft stubs run in two 608 bearings in printed pillow blocks at $y = \pm 50$ mm, and are grub-screwed into two printed crank webs (44 mm diameter) that carry a 5 mm crank pin at radius $r = 13$ mm. The connecting rod (17) is a fork: its lower eye runs on the pin *between* the two webs, and its two upper plates straddle the arm and pivot on a 5 mm pin at A, 60 mm from the arm pivot. Running the rod between the webs rather than beside a continuous shaft is what lets the pin pass below the shaft axis at the bottom of the stroke without the rod striking the shaft; this was a finding of the interference audit (Sec. 3.8).

The crank centre $C = (60,\ 23.8)$ mm is placed exactly one connecting-rod length $\ell = 71.2$ mm below A's mid-stroke position, so the two dead centres of the crank (pin directly above and below C) put A at $z = 95 \pm 13$ mm and the swing is symmetric:

$$\sin\theta_{\max} = \frac{r}{r_{\mathrm{con}}} = \frac{13}{60} \quad \Rightarrow \quad \theta = \pm 12.5^{\circ}, \qquad s = 2L_{\mathrm{arm}} \sin\theta_{\max} = 54.2\ \mathrm{mm}. \tag{1}$$

For a general crank angle φ the arm angle follows from loop closure $|A(\theta) - B(\varphi)| = \ell$ with $A = r_{\mathrm{con}}(\cos\theta, \sin\theta) + P$ and $B = C + r(\cos\varphi, \sin\varphi)$, solved numerically (Fig. 7b). The arm is horizontal when $\varphi_0 = \arcsin(r/2\ell) = 5.2^{\circ}$, the rest pose used in the CAD. At 60 rpm the block face crosses the blade at 153–174 mm/s and the 20 mm face is cut in 118 ms (Fig. 7c), i.e. the cutting-force pulse has its energy below ∼9 Hz, far from the first structural mode (357 Hz).

The torque the handwheel must supply is small: the pivot's elastic restoring torque peaks at 0.058 N m, which through the 60 mm arm joint, the rod and the 13 mm crank becomes ≈ 0.013 N m at the shaft, or 0.23 N at the handle, imperceptible against the cutting force itself (0.6–2.4 N along the stroke).

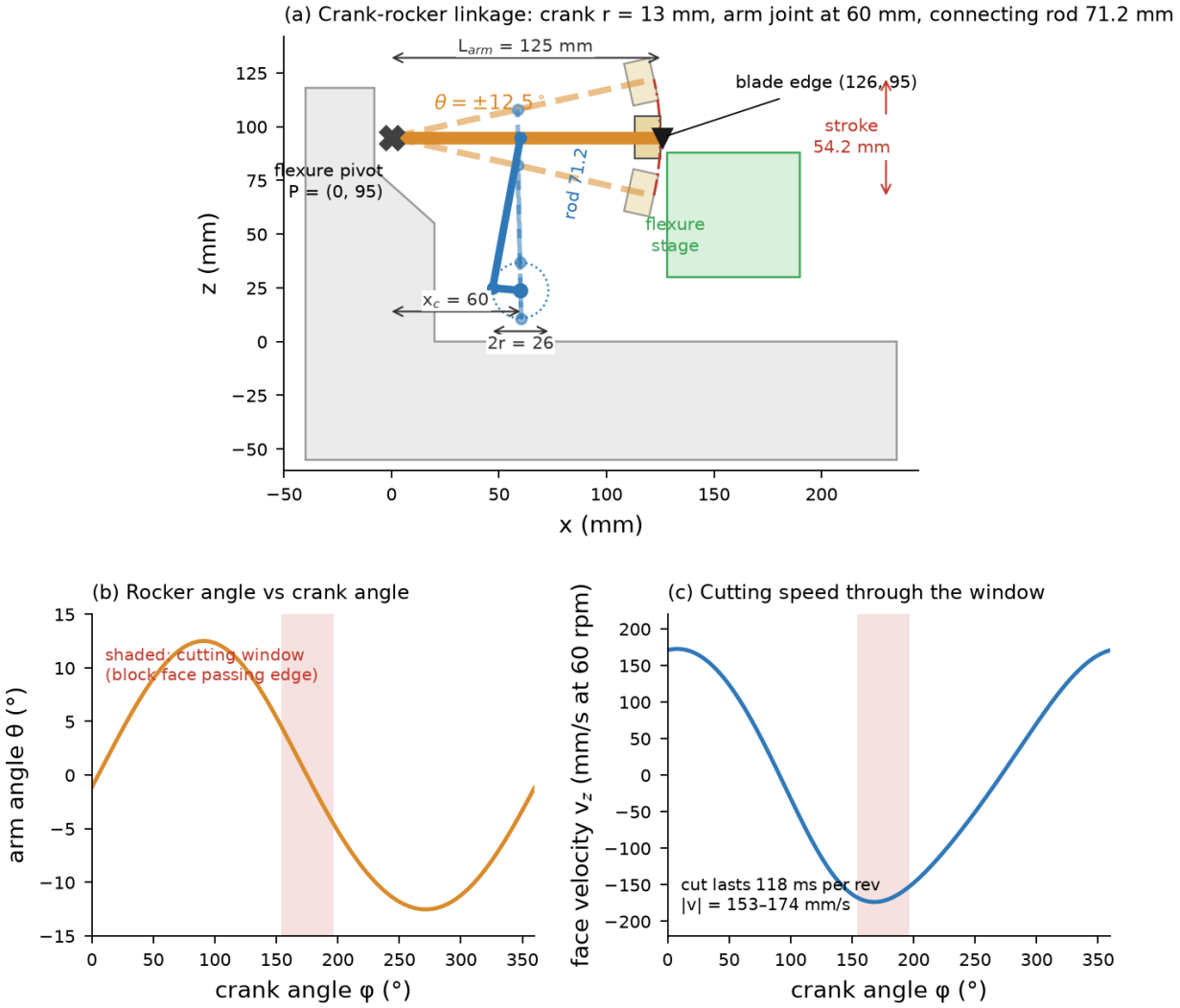


Figure 7: Crank-rocker kinematics. (a) Linkage to scale at mid-stroke (solid) and at the two extremes (dashed), with the block on the arm tip, the face arc (red dash-dot) and the stationary blade edge. (b) Arm angle θ against crank angle φ; the shaded band is the cutting window, during which the 20 mm face passes the edge ($|\theta| < 4.6°$ on the down-stroke). (c) Vertical face velocity at 60 rpm.

3.2 The cut: blade and block geometry

The blade (6) is a standard disposable low-profile blade ($80 \times 8 \times 0.25$ mm, 35° included facet angle). Its rear face rests against a full-width printed back wall on the holder (7), which is what carries the cutting force (the force pushes the blade away from the specimen, onto the wall); a 1.0 mm steel clamp strip in front of the blade is pulled flat onto a 0.20 mm blade recess by two M4 screws below the blade, so that the 0.25 mm blade is pinched by 0.05 mm of strip interference. The edge stands 2.5 mm proud of both the strip and the wall. The holder leans the blade back 27° from the vertical block face. With a 35° wedge the three angles at the edge (Fig. 8a) are therefore

$$\alpha_{\text{clearance}} = 27° - \tfrac{35°}{2} = 9.5°, \qquad \gamma_{\text{rake}} = 90° - 27° - \tfrac{35°}{2} = 45.5°, \qquad \alpha + \tfrac{\beta}{2} + \gamma = 90°. \quad (2)$$

Because the block rides an arc of radius $L_{\text{arm}} = 125$ mm, the face tangent rotates with θ during the cut, and the instantaneous clearance sweeps $9.5 \pm 4.6°$ (4.9–14.1°) across the face, inside the 3–15° window practitioners recommend [14, 18]. The same arc means the cut face is cylindrical, with sagitta

$$h = L_{\text{arm}} - \sqrt{L_{\text{arm}}^2 - (b/2)^2} = 401\,\mu\text{m} \quad (b = 20\,\text{mm}), \quad (3)$$

but since every stroke follows the same arc about the same (flexure-defined) centre, the arc is trimmed into the block once and successive sections (differences between successive arcs) are of uniform thickness (Fig. 8b). The strip top sits 2.5 mm below the edge along the blade, which places it just outside the block's swept circle; the back wall, being behind the blade, is not bound by that circle and could rise to 2.0 mm (Sec. 5.6).

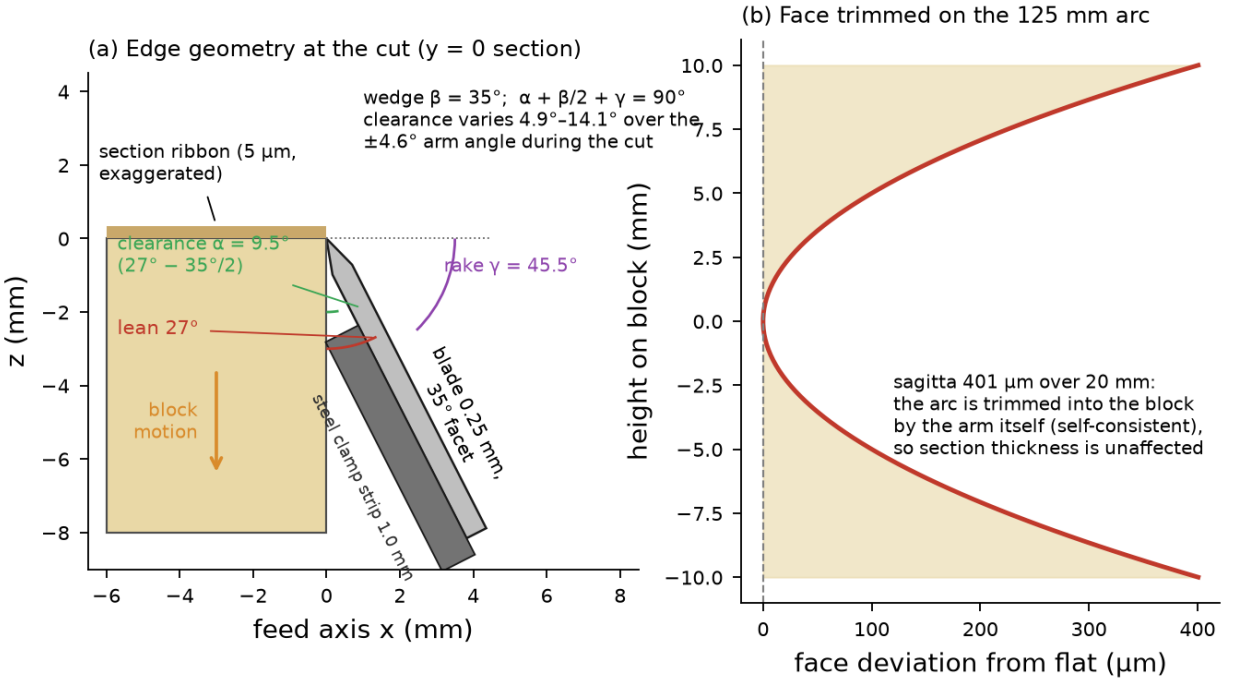


Figure 8: Edge geometry. (a) Mid-plane section at the cut: block (moving down), blade leaning 27°, steel clamp strip, and the clearance, wedge and rake angles. (b) Deviation of the arc-trimmed face from a plane over a 20 mm block.

3.3 The arm pivot: a cross-strip flexure

The arm (3) is not on a pin or a bearing. Its root is held to the tower by three strips of 0.20 mm feeler-gauge steel (2), two in one plane (10 mm wide each, at $y = \pm 15$ mm) and one in the perpendicular plane (20 mm wide, at $y = 0$), crossing at the pivot axis with 20 mm free length and clamped at both ends under printed pads with M3 screws (Fig. 11a). The arm rotates by bending the strips; there is no sliding surface, so there is no backlash, no lubrication and no wear, and the restoring torque is elastic:

$$k_\theta = \sum_i \frac{E\, w_i t^3}{12\, L} = 0.267\,\mathrm{N\,m/rad}, \qquad \sigma_{\max} = \frac{E\, t\, \theta_{\max}}{2L} = 218\,\mathrm{MPa}, \qquad T_{\max} = k_\theta \theta_{\max} = 0.058\,\mathrm{N\,m}. \tag{4}$$

The stress is a factor 2 below the fatigue limit of hardened spring strip and 5 below yield. Along the feed axis the pivot is loaded in tension/compression of the strips and is very stiff (40 N/µm; 0.025 µm/N in the loop). The price of a cross-strip pivot is a second-order shift of the rotation centre, $\approx 0.1\, L\theta^2 \approx 95$ µm at full swing [12, 13, 17]; because it is a deterministic function of θ it reshapes the arc slightly but is identical stroke to stroke, so it does not enter the thickness budget. Making the middle strip twice as wide as the outer two and placing it between them balances the two strip planes about $y = 0$, so the pivot has no net twist about z when the arm is loaded off-centre.

3.4 The feed train: differential screw, ratchet drum and selector

The feed moves the blade, not the specimen. The blade stage (8) is a monolithic parallelogram: a platform on two 1.0 mm PLA plates (front plate 60 mm wide, rear pair 2×20 mm) 44 mm tall, so the platform translates along x with no sliding joint. A gusseted tab hanging from the platform carries an M8×1.0 nut on a steel washer; a 10 mm thick, 64 mm wide upstand, continuous with the stage foot and braced by a central and two side gussets, carries an M8×1.25 nut on a washer. One screw assembly (11) threads through both: an M8×1.0 hex bolt in the tab, a 13 mm printed coupler that holds the bolt head in a hex pocket and an M8×1.25 thin nut in a second pocket, and the M8×1.25 rod, screwed through that nut until its end bears on the bolt head and locked with thread locker; the rod continues through the frame nut to the ratchet drum (12), two jam nuts and the feed knob (14). The printed coupler therefore carries only torque; the axial force path from blade to frame is steel throughout. Turning the screw one revolution advances it 1.25 mm through the frame nut, while the tab nut retreats 1.00 mm along the screw (Fig. 9a); the stage therefore moves by the difference,

$$\Delta x_{\mathrm{rev}} = p_1 - p_2 = 1.25 - 1.00 = 0.25\,\mathrm{mm}, \qquad \Delta x_{\mathrm{tooth}} = \frac{\Delta x_{\mathrm{rev}}}{50} = 5.0\,\mu\mathrm{m}. \tag{5}$$

Two extension springs between the tab and the upstand ($\sim$10 N total) pull the tab in $+x$, the same direction as the cutting force, so the nut always rests on one thread flank and lead-screw backlash is removed; the symmetric flexure's own restoring force (4.9 N at full travel) would reverse at mid-travel and is overridden by the springs. Three consequences follow from the differential principle. (i) *Reduction*: the screw body translates five times farther than the stage, so the 14 mm drum, not a thin wheel, keeps the pawl engaged over the ±5 mm screw excursion that accompanies the ±1 mm stage travel (400 sections at 5 µm per reset). (ii) *Tolerance*: only the *difference* of the two pitches matters, but that difference is a fifth of either pitch, so relative lead errors are amplified five-fold: $\delta t/t = (\delta p_1 - \delta p_2)/0.25\,\mathrm{mm}$. With a cumulative lead error within ±0.1 % on each thread this is a per-machine bias within ±0.7 %, removable by one calibration; the periodic and local lead errors add 0.18 µm (1 s.d.) of scatter at 5 µm. (iii) *Resolution without precision parts*: both threads are ordinary rolled hardware; the micrometre-scale motion lives in steel while the printed parts only have to hold nuts.

The thickness selector (13) is a sector shield on the stage column that masks all but $k = 1 \ldots 6$ teeth from the pawl finger, so each stroke advances $5k$ µm (Fig. 9b). The coarse approach after loading a new block is independent of the fine feed: the stage foot is clamped to the deck through ±10 mm slots by two M6 knobs (9) and is pushed against an M6 screw in a side boss (10).

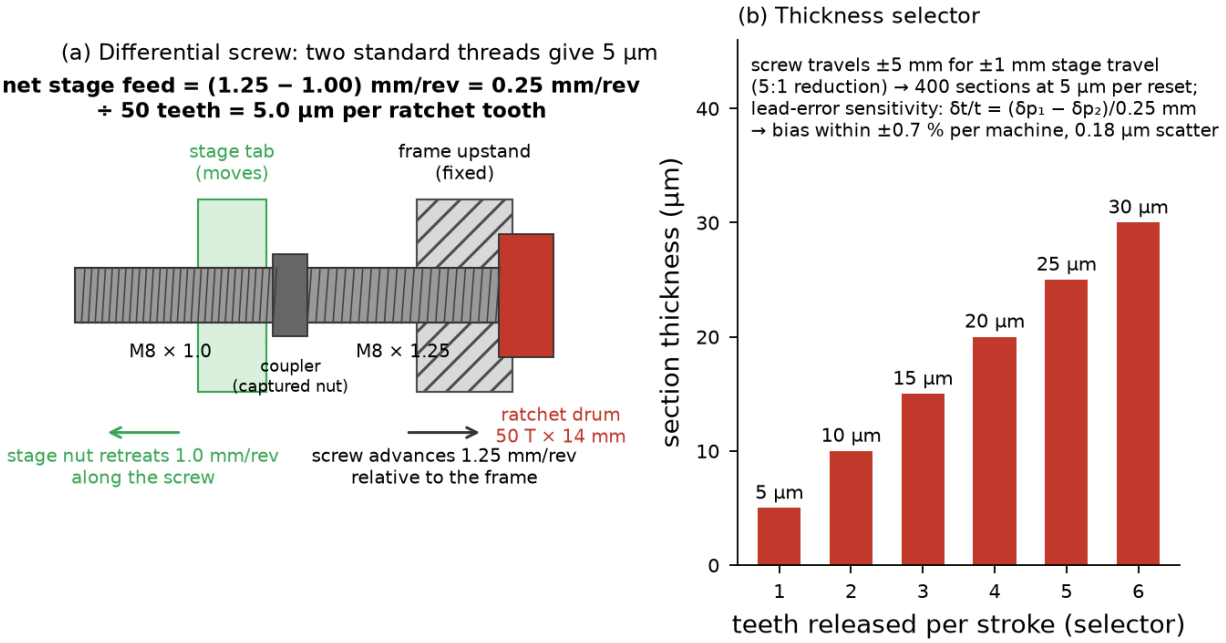


Figure 9: Fine feed. (a) Differential screw: the frame nut (M8×1.25) and the stage-tab nut (M8×1.0) move the stage by the pitch difference, 0.25 mm per revolution, divided into 50 ratchet teeth. (b) Selector settings and the resulting nominal section thickness.

3.5 Feed actuation: how one turn makes one section

A vertical pawl rod (15) slides in a guide on the stage column; its finger rests on the drum teeth under gravity and a light spring. A striker bar (16) fixed to the arm reaches over the rod's flag. On the up-stroke, once the arm passes $\theta \approx 6^\circ$ (block 13 mm clear of the edge) the bar lifts the rod, the finger slides back over the exposed teeth, and at the top of the stroke ($\theta = 12.5^\circ$, lift 22 mm) it has ridden over k teeth. As the arm comes back down past $\theta \approx 6^\circ$ the bar releases the rod; the finger drops behind the kth tooth and the rod's weight and spring turn the drum by $k \times 7.2^\circ$, advancing the blade $5k$ µm toward the specimen *before* the face reaches the edge (Fig. 10). The feed is therefore complete and the drum is at rest during the cut; the ratchet never moves while the blade is in the block. Return-stroke retraction of the specimen (lifting it off the blade on the up-stroke) is not implemented; the arc geometry gives 3.0 mm of natural retraction at the stroke ends but none at mid-face, so the back of the blade brushes the fresh face on the up-stroke as in the classical Cambridge rocker [16].

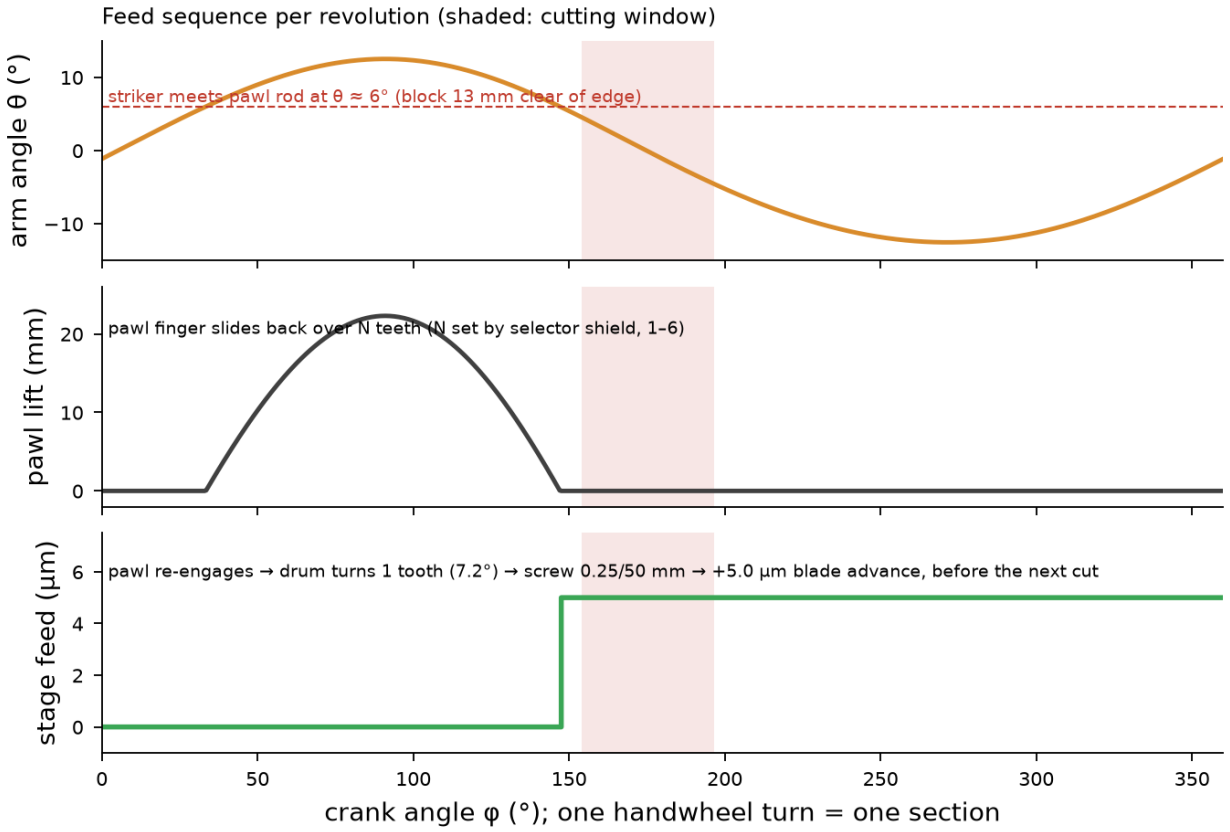


Figure 10: Timing over one handwheel revolution: arm angle, pawl-rod lift and stage feed. The feed step occurs on the down-stroke before the cutting window (shaded).

3.6 The force path and why most of the machine does not matter

Section thickness is set by the distance between the blade edge and the block face at the moment of the cut. Only parts on the closed path between those two points can change it; everything else (handwheel, crank, rod, drum, pawl) can be as flexible as PLA allows. Walking that loop in the section view (Fig. 6): edge → blade overhang → back wall and holder → platform → flexure plates (pitching about the tab nut) → tab nut on its washer → bolt, rod end on bolt head, rod → frame nut on its washer → gusseted upstand → foot, deck, base → tower → strip pivot → arm → cassette clamp → block. The feed-axis compliances add in series (Fig. 11b):

$$C_{\text{loop}} = \sum_i C_i = 1.20\,\mu\text{m/N}, \qquad k_{\text{loop}} = 1/C_{\text{loop}} = 0.83\,\text{N}/\mu\text{m}. \tag{6}$$

A steady 2 N feed-axis cutting force thus opens the loop by 2.4 µm, but a steady deflection is cut into the block face and does not change the thickness; only its stroke-to-stroke *change* does (0.4 N per-stroke s.d., hence $\sqrt{2}\times 0.4$ N of change → 0.68 µm). The three largest terms are the frame on two end pads (0.35, FE), the blade overhang (0.25) and the gusseted upstand (0.19, FE); in revision 1 the ungusseted upstand alone was 7.3 and dominated everything, which is why the external review of that release was decisive (Sec. 3.9).

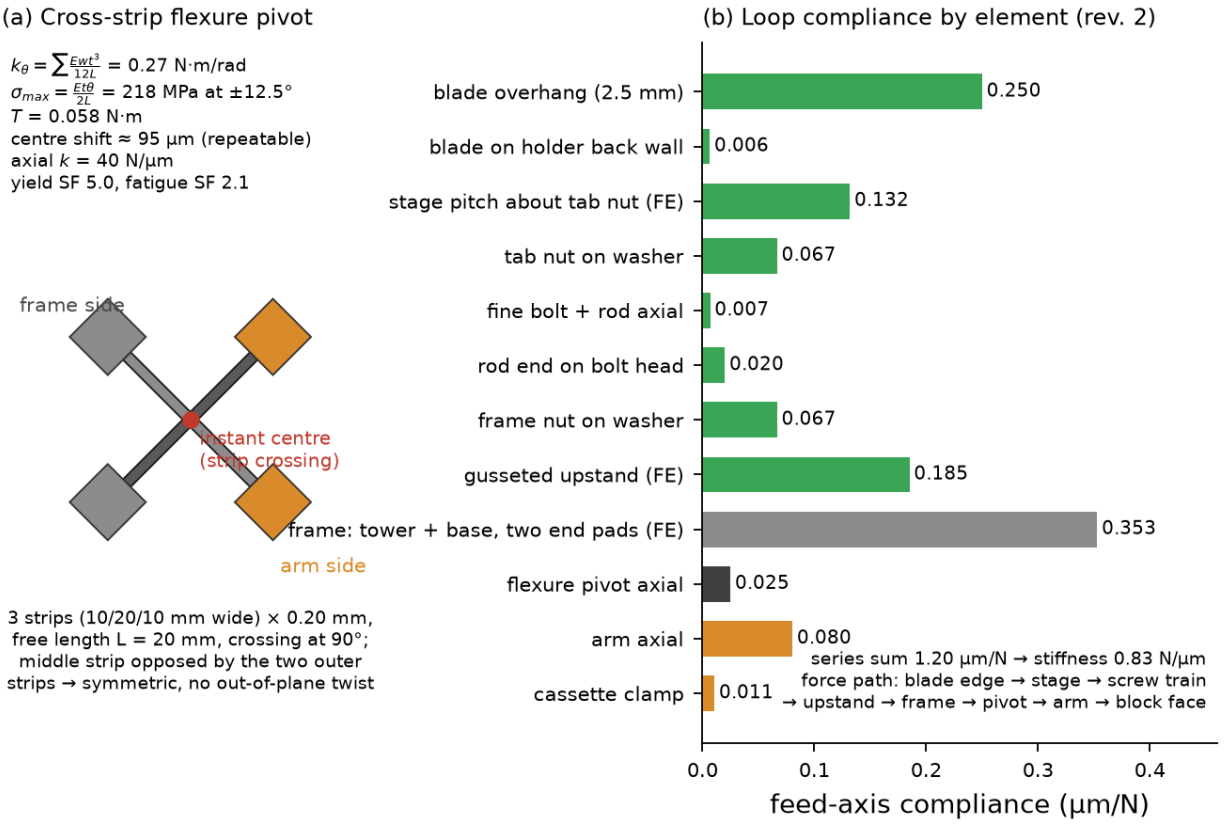


Figure 11: (a) Cross-strip pivot geometry and governing relations. (b) Feed-axis compliance of each element of the metrology loop (green: blade stage group; grey: frame; dark: pivot; orange: arm and clamp).

3.7 Operating sequence

1. Clamp the cassette under the sliding jaw with the M5 thumbscrew (4); the block face must stand proud of the jaw.
2. Loosen the two M6 knobs (9), push the stage toward the block with the coarse screw (10) until the edge is ∼0.5 mm from the face, re-clamp.
3. Set the selector (13) to 6 teeth (30 µm) and turn the handwheel until the whole face is trimmed to the arc.
4. Set the selector to 1 tooth (5 µm); each further turn yields one 5 µm section, which the previous section's wax edge carries off as a ribbon.
5. After ∼400 sections the stage reaches the end of its ±1 mm travel: wind the feed knob (14) back 20 turns and repeat step 2.

3.8 Design revisions driven by the interference audit

A boolean interference audit of the assembly (`audit_interference.py`: 12 crank angles over the revolution × 3 feed-screw positions at −5, 0, +5 mm, all part pairs) and a mesh-connectivity check (every exported STL must be exactly as many connected shells as its quantity) were re-run after every geometry change. Over the two revisions they caught, and the design now reflects: the blade holder entering the specimen's swept circle (27° lean and 2.5 mm overhang); the striker bar passing through the pawl-rod body (bar moved outboard, flag added); the coarse-approach boss fouling the stage foot; the connecting-rod eye at the wrong crank position; the screw body translating five times the stage travel (drum ratchet, coupler gap, shield on the column); the connecting rod sweeping through a continuous crankshaft at the bottom of the stroke (built-up two-web crankshaft); a 1.2 mm offset of the crank centre that made the swing asymmetric (−11.4° to +13.7°; centre lowered to $z = 23.8$ mm for ±12.5°); and, in revision 2, the drum striking the thicker upstand at −5 mm

(drum, shield and knob shifted 3 mm), the coupler and bolt head passing through the lowered platform (screw axis lowered to $z = 60$ mm), the knob heads touching the new side gussets, the clamp-screw heads below the platform top (relief pockets), and enclosed pivot pockets and a sealed arm cavity in the meshes. Of these, five were literal collisions in the posed assembly and the rest were geometry or kinematic errors found by inspecting the same poses. The residual intersections are intended and bounded in the script: pins in bores, strips in clamp pockets, the pawl tooth in the drum (16 mm^3), spring hooks on their pins, and the striker bar on the pawl flag between 6° and 12.5° of arm angle, which is the actuation itself.

3.9 Load-path corrections after external review (revision 2)

The first release of this design was reviewed independently against its own CAD and code. The review found that the structural-loop budget did not describe the load path the CAD actually contained, and the design and model were revised accordingly. The findings and corrections are listed here because they changed the headline numbers (Table 2).

Table 2: External-review findings on revision 1 and the revision-2 response.

Finding (revision 1)	Why it mattered	Revision 2
Frame-nut upstand not connected to the stage foot (2 mm gap) and only 6×32 mm; omitted from the budget	A 7.3 µm/N cantilever in series with everything else: the loop would have been ≈8.5 µm/N and the scatter ≈4.8 µm	Upstand continuous with the foot, 10 mm wall, 64 mm wide, central and side gussets: 0.185 µm/N by FE (Sec. 4)
Blade held between a front pressure strip and two rear grub-screw tips	The cutting force pushes the blade *away* from the strip, onto two screw points; the budget's full-width compression term did not represent this	Full-width printed back wall behind the blade carries the force; the strip clamps the blade onto a 0.20 mm recess with two M4 screws below the blade
Coupler 0.7 mm thick behind the bolt head; coarse rod not engaged with it	Unsubstantiated strength and stiffness of the only link between the two threads	13 mm coupler with a captured M8×1.25 nut; the rod is screwed in until its end bears on the bolt head, so the axial path is steel on steel
Budget contained a 6 mm ball contact that exists nowhere in the CAD; only one nut-bedding term	Fictitious and missing terms	Ball term removed; tab-nut and frame-nut bedding both included, on steel washers; coupler contact added
Flexure preload reverses across mid-travel	Nut could change thread flank mid-ribbon	Two extension springs preload the tab in the direction of the cutting force
Cutting-force variation applied as $C\,F_i$	Thickness depends on $F_i - F_{i-1}$: the force term was $\sqrt{2}$ too small	Monte-Carlo rewritten so every error enters as a stroke-to-stroke difference
Thread pitch redrawn independently every stroke	Real lead error is fixed along the screw and correlated step to step; it produces bias as well as scatter	Lead error modelled as cumulative + periodic + local random walk per simulated screw pair; bias reported separately
Frame FE supported along its whole underside	A warped print bears on its ends	Two-end-pad case computed and adopted
Disconnected mesh shells (stage, selector strap), missing M3 holes, sealed arm cavity; audit script absent	Unprintable or unassemblable as exported; audit not reproducible	All 35 STLs now single connected shells (checked in `audit_interference.py`, which is in the repository); insert holes added; arm cavity vented
No scripts for the modal, transient, fracture and BOM numbers	Not reproducible	`extra_analysis.py` and `make_bom.py` added

The revision lowered the deck by 8 mm (to $z = 22$ mm) so that the taller blade holder fits above the platform while the flexure plates keep their 44 mm free height, lowered the screw axis to $z = 60$ mm, and shifted the drum, shield and knob 3 mm outward to clear the thicker upstand. With the connected load path the loop compliance is 1.20 µm/N; had revision 1 been built as drawn, the model predicts it would have cut 5 ± 4.8 µm.

4 Analysis methods

Structural-loop model. Each element between the blade edge and the specimen face was assigned a compliance along x under a unit cutting force: blade overhang bending and bearing on the back wall, the FE-calibrated stage (pitch about the tab nut), tab-nut bedding, the screw column and the rod-on-bolt-head contact, frame-nut bedding, the FE-calibrated upstand, the FE-calibrated frame (tower + base), the pivot strips in tension, the arm in compression, and the cassette clamp. Nut bedding on a 16 mm steel washer is taken as 1.5×10^7 N/m (0.5×10^7 N/m for a bare nut on PLA layer lines; both are allowances, not measurements). PLA $E = 3.5$ GPa [10], steel 200 GPa.

Monte-Carlo thickness budget. Section i is the difference between the blade position during cut i and during cut $i - 1$, so every error source enters as a stroke-to-stroke difference. 200 screw pairs × 100 consecutive strokes were simulated with: a lead-error function fixed along each thread (cumulative error uniform within ±0.1 %, per-revolution periodic error of 0.75–1.5 µm amplitude, and a local random walk of 0.3 % of pitch per step), evaluated where each nut sits at each stroke; ratchet tooth-pitch error of 0.05 mm (1 s.d.) on the 50 mm pitch circle; per-stroke cutting-force variation of 0.4 N (1 s.d.), so that the force *difference* between strokes has s.d. $\sqrt{2} \times 0.4$ N, acting through the loop compliance; thermal drift over the 1 s stroke interval at 1 K per 30 min room drift; and 0.1 µm pawl seating. The cumulative lead term appears as a per-machine bias, reported separately from the scatter. The nominal cutting force (2 N) and its variation are assumptions (no absolute paraffin cutting-force measurement exists in the literature [18]), and Fig. 3b shows the sensitivity.

Finite elements. 2-D plane-stress linear elasticity (scikit-fem, quadratic triangles) of the stage cross-section (load at the blade, foot and tab nut fixed), of the stage foot with the frame-nut upstand (load at the nut-pocket floor, foot bottom fixed; run for the revision-1 and revision-2 geometries) and of the frame side-section (self-equilibrated pivot-to-deck couple; underside fixed either fully or on two end pads), with out-of-plane widths represented by scaling E per region. Verified against a cantilever benchmark (tip deflection 97.3 mm/N analytic vs 97.3 FE).

Modal estimates. Lumped single-degree-of-freedom estimates for arm bending in the swing plane, stage x-motion and whole-loop x-motion.

Cost. Printed-part cost from solid-modelled volume at US$20/kg PLA (an upper bound; ribbed and infilled prints weigh less); hardware at typical online/hardware-store prices.

5 Results

5.1 Stiffness budget

Total feed-axis compliance is 1.20 µm/N (stiffness 0.83 N/µm). The three largest terms are the frame on two end pads (0.35 µm/N, FE), blade overhang (0.25) and the gusseted upstand (0.19, FE); the stage pitch is 0.13, the two nut seats 0.07 each, and the pivot strips only 0.02 (Fig. 12a). With the revision-1 upstand and clamp as drawn the total would have been 8.5 µm/N.

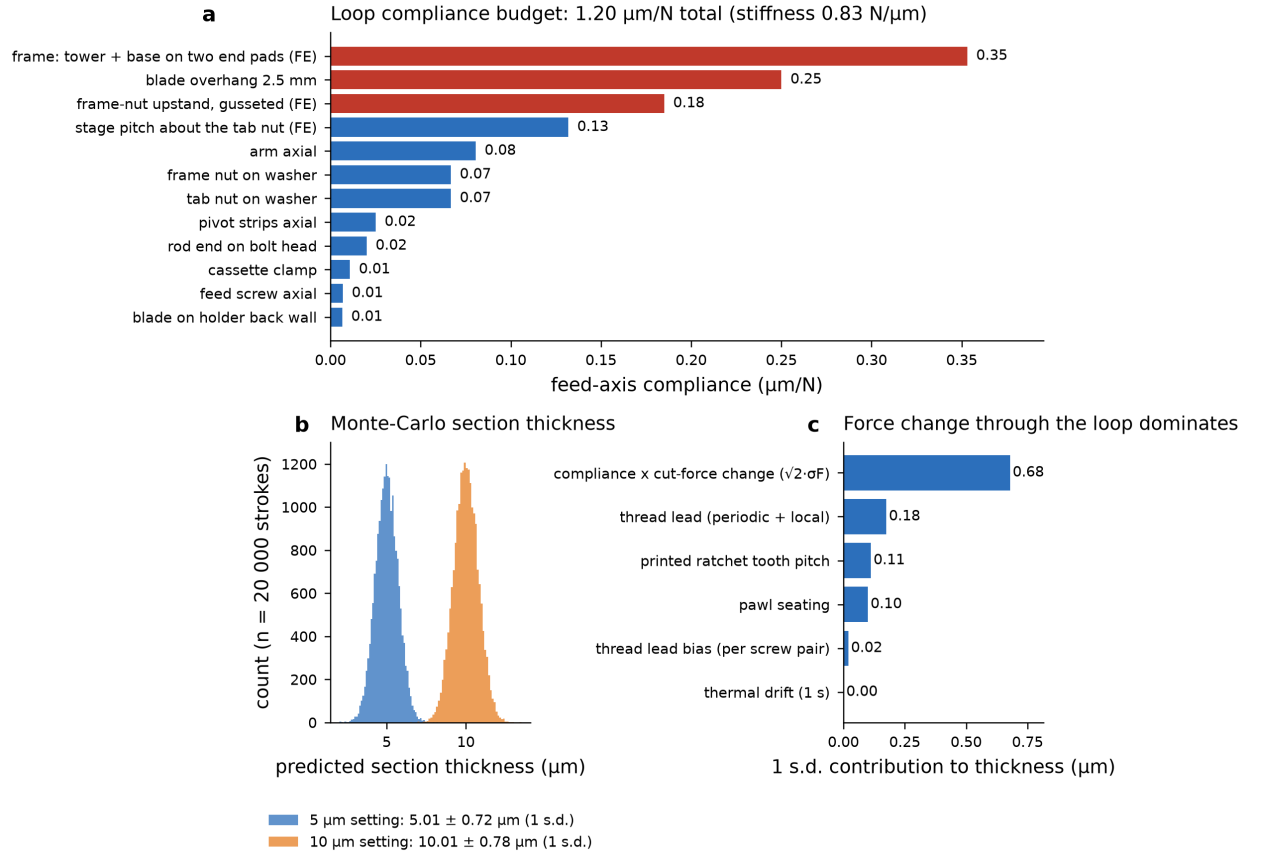


Figure 12: (a) Compliance budget of the metrology loop. (b) Monte-Carlo section thickness at the 5 µm and 10 µm settings ($n = 20\,000$). (c) 1-s.d. contribution of each error source at the 5 µm setting.

5.2 Predicted section thickness

At the 5 µm setting the predicted section is $\mathbf{5.01 \pm 0.72}$ µm (1 s.d.), 90 % interval 3.84–6.19 µm; at 10 µm, 10.01 ± 0.78 µm. The stroke-to-stroke change in cutting force acting through the loop compliance (0.68 µm) dominates; thread lead (0.18 µm), printed-ratchet error (0.11) and pawl seating (0.10) follow; thermal drift within a ribbon is negligible (Fig. 12c). The cumulative lead error of the two threads adds a per-machine bias of ±0.7 % (90 % interval), which a one-time calibration against a measured ribbon removes. For context, direct measurement at 154 sites found more than two-thirds of sections from commercial rotary microtomes outside ±10 % of the dial setting [19]; FlexTome's predicted 1-s.d. scatter is ±14 %.

5.3 Thermal behaviour

The PLA loop expands 10.6 µm/K. This is irrelevant within a ribbon (strokes 1 s apart) but means the first sections after a temperature change or after mounting a new block will run thick or thin by several micrometres until trimmed, the same behaviour the Leica microtomy guide describes for a cold block, and handled by the same routine (trim, then section).

5.4 Dynamics and chatter

Estimated first modes: arm bending 357 Hz, whole-loop feed-axis 459 Hz, blade stage 1.5 kHz. A 20 mm face is cut in 118 ms, so the cutting excitation is concentrated below ∼9 Hz; no resonance-driven chatter is predicted. The rotational mode of the arm on its pivot (≈4 Hz) is the actuated degree of freedom and is constrained by the connecting rod.

5.5 Cost and manufacturability

Table 3: Bill of materials (US$).

Category	Cost
16 printed part types (20 pieces), 1.40 L PLA (solid-modelled upper bound)	34.57
Hardware (feeler gauges, bearings, shaft, rods, nuts, washers, screws, springs)	24.80
Consumables (one blade, adhesive, grease)	2.60
Total	**61.97**

Against US$8 995–22 690 new and US$2 100–7 950 used, this is a $> 99\,\%$ and $> 97\,\%$ reduction respectively. The largest part (frame, $235 \times 150 \times 173$ mm) fits an Ender-3-class 235 mm bed (or is split at the deck step for smaller printers); all other parts fit a 180 mm bed. No machining is required; the only cutting operations are shearing feeler-gauge strips to length and cutting threaded rod.

5.6 Cutting-force bound, improvement levers and the entry transient

Because no absolute paraffin cutting force has been published, the 2 ± 0.4 N assumption was bracketed with the pressing-and-slicing fracture model [20]: with a fracture work $R = 20$–$80\,\mathrm{J/m^2}$, wax yield stress 1–3 MPa, a 35° facet and friction coefficient 0.3, a 5 µm × 25 mm cut requires 0.6–2.4 N along the stroke and only 0.23–0.86 N along the feed axis ($\leq$1.3 N for a dulled blade with doubled friction). The feed-axis force that drives thickness error is therefore likely below the 2 N assumed, and the 0.4 N (1 s.d.) variation is conservative; at 0.2 N the predicted scatter falls to 0.41 µm.

Three levers remain. (i) The blade overhang on the *back-wall* side is not limited by the specimen's swept circle (that limit applies to the front strip, whose top would enter the circle by 0.09 mm at 2.0 mm), so the wall top can rise to 2.0 mm below the edge with a relief chamfer for the section: 1.20 →1.08 µm/N, scatter 0.72 →0.65 µm. (ii) Bedding the base fully on the bench (or shimming its ends) recovers the full-contact frame value: 1.20 →1.06 µm/N, 0.64 µm. (iii) Brass inserts in place of washers under the two nuts: 1.20 →1.17 µm/N. All three together give 0.90 µm/N and 0.56 µm (Fig. 3b). Bare nuts on PLA, by contrast, would give 1.47 µm/N and 0.86 µm.

Within a single section, the cutting force rises over the ∼0.5 mm entry chamfer of the block face in ∼3 ms. Driving the 459 Hz, 3 %-damped loop with that ramp gives a 17 % dynamic overshoot (0.41 µm on a 2.4 µm quasi-static deflection) that rings down within 46 ms of the 118 ms cut, bounding within-section thickness non-uniformity from this source to ≈0.4 µm over the first 7 mm of the face.

6 Discussion

What the analysis establishes. The predicted 5.0 ± 0.7 µm is a model result, not a measurement. Its value is that it identifies, quantitatively, what limits a printed microtome: not feed resolution (the steel differential screw delivers 5 µm trivially) but loop stiffness against stroke-to-stroke cutting-force change, and within that, frame support, blade overhang and the frame-nut upstand. Two lessons about method stand out. The frame FE showed that a beam-theory budget built on an idealised tower section under-predicted tower compliance eight-fold and credited the tie rods with a stiffening they do not provide. The external review of revision 1 showed something more basic: a budget is only as good as its correspondence to the drawn load path, and the drawn load path had a 7.3 µm/N element the budget never contained. Checking that every element in the force path is a connected, fastened body in the CAD, and that every term in the budget is one of those bodies, is now part of the release checklist (Sec. 3.9).

Relation to prior art. The Darwin rocker already combined arm, screw and ratchet; what it lacked was a bearing that a printed machine could reproduce. Replacing its hardened steel knife-edge or journal pivot with crossed feeler-gauge strips removes the one component that required a machine shop. The differential screw is a textbook element [13] but has not previously been reported in a microtome, where it allows the 5 µm increment to come from two hardware-store threads rather than a micrometer spindle. The parallelogram blade stage follows the OpenFlexure precedent that FDM compliant mechanisms achieve micron-scale repeatability [21].

Limitations and risks.

1. *Cutting force is assumed.* The 2 ± 0.4 N figure is an engineering estimate bracketed by fracture mechanics; no measured value exists. Doubling the variation to 0.8 N raises scatter to 1.04 µm. A load-cell measurement on the build is therefore both a validation step and a contribution in its own right.
2. *No specimen retraction on the return stroke.* The face passes the edge on the upstroke flush with it. Basic manual rotary microtomes without retraction are used routinely, but knife lines are possible; a 50–100 µm retraction cam acting on the stage is a planned addition.
3. *Cylindrical block face.* The 0.40 mm sagitta over 20 mm is the rocker's known property; sections flatten on the water bath. Block height is limited to ∼25 mm by the clearance-angle variation.
4. *PLA creep and humidity.* The stage flexure is loaded only while displaced from neutral ($\leq$5 N, 5.4 MPa); the user returns the screw to centre after each block. PETG/ASA or 100 % infill at 0.1 mm layers reduce creep further [9].
5. *Thread backlash.* The flexure preload and the unidirectional cutting load keep the nut flanks engaged; reversing the feed requires re-trimming, as on any manual microtome.
6. *Blade clamping.* A printed back wall and a 1 mm steel clamp strip with two screws is simpler than a commercial clamp; blade seating repeatability and the flatness of the printed wall (layer lines run along the blade) must be verified.

Broader impact. A $62 microtome does not create a pathology service; it needs embedding, staining, a microscope and, above all, a pathologist or a telepathology link. But it removes the single highest-cost mechanical item from the chain, and does so with a device that a hospital workshop or a school with a desktop printer can build and repair. Combined with the OpenFlexure microscope, it brings the entire physical chain from block to image under US$300.

7 Validation plan (for the physical build)

1. **Print and assemble** (frame 100 % infill in the pivot and deck regions, 0.2 mm layers; stage flexure 100 % infill, 0.1 mm layers, plates printed vertically). Record print mass and time.
2. **Static loop stiffness.** Apply 0.5–3 N to the block face along x with a hanging mass over a pulley; read displacement with a 1 µm dial indicator at the blade edge. Target $\geq$0.8 N/µm. Compare to the budget element by element by moving the indicator.
3. **Feed calibration.** Advance 50 teeth (one revolution) and measure 250 ± 5 µm stage travel; repeat 10× for pitch-error scatter.
4. **Section thickness.** (a) Block-consumption method: dial indicator on the trimmed face before and after 100 sections at 5 µm. (b) Gravimetric: weigh ribbons of 20 sections of known face area (paraffin density 0.90 g/cm^3). (c) Optical: measure 20 individual sections with a confocal/interference microscope if available, following Fujisawa et al. [19]. Report mean, s.d. and the fraction within ±10 %.
5. **Cutting force.** Mount a 5 N load cell under the stage foot; record force vs crank angle over 50 strokes to replace the assumed 2 ± 0.4 N.
6. **Section quality.** Cut 5 µm ribbons from standard test blocks (liver, kidney, skin) on FlexTome and a commercial microtome; H&E stain; blinded scoring (compression, chatter, knife lines, folds) by two histotechnologists.

7. **Durability.** 10 000 strokes (≈3 h at 60 rpm, motor-driven), then repeat steps 2–4.

8 Conclusion

A rocking microtome whose precision-critical joints are flexures and whose micrometre feed chain is entirely steel can be built from printed PLA and hardware-store parts for US$61.97. The FE-checked design analysis, corrected after external review of its load path, predicts 5.0 ± 0.7 µm sections at a loop stiffness of 0.83 N/µm, with stiffness against stroke-to-stroke force change, not feed resolution, as the limiting quantity, and identifies base support, blade overhang and the frame-nut upstand as the levers for improvement. The prediction has not been tested on a physical build; that is the next step. The complete CAD, analysis code and validation protocol are released with this report so that the prediction can be tested, and the design iterated, by anyone with a desktop printer.

Data and code availability

`flextome_assembly.step` (full assembly), `flextome_stl_parts.zip` (30 component STLs + `parts.json`), `flextome_cad.py` (parametric CadQuery source), `flextome_design.py` (analytic model and Monte-Carlo), `fea_stage.py`, `fea_frame.py` (finite-element models), `make_figures.py`, `flextome_bom.csv`, `references.csv` and `evidence_summary.md` (verified literature base).

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